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Expert Answer

General Guidance

Equilibrium of a rigid body:

An object is said to be in equilibrium when the sum of external forces and couples are zero.

For a rigid body to be in equilibrium in three dimensions, the sum of external forces acting along

$$x$$

,

$$y$$

and

$$z$$

directions have to zero.

Tension:

The tension force is the force that is transmitted through a chain, string, rope, cable, wire or similar one dimensional object when it is pulled by forces acting from opposite ends.

Friction force:

It refers to the force that opposes the relative motion between sliding bodies.

Normal reactions:

Normal reactions are the forces which act in the direction perpendicular to the supports of the system.

Coefficient of static friction:

It is referred as the ratio of the frictional force acting at the body to its normal force. It is denoted by

$$\mu$$

and it is dimensionless variable.

Apply the principle of equilibrium along the y direction to find out the normal force exerted on the crate A . To find out the tension in the cable, calculate the friction force in the crate A using the friction force formula and apply the principle of equilibrium along the x direction on the cart A to find out the tension in the cable.

Similarly apply the principle of equilibrium along the y direction to find out the expression for the normal force exerted on the crate B . Calculate the friction force in the crate B using the friction force formula and apply the principle of equilibrium along the x direction on the cart B to find out the maximum force that can be applied without causing the two crates to move.

Write the equilibrium of forces along the

x

-axis.

$$\sum F_x = 0$$

Here, sum of forces along x -direction is

$$\sum F_x$$

.

Write the equilibrium of forces along the

y

-axis.

$$\sum F_y = 0$$

Here, sum of forces along y -direction is

$$\sum F_y$$

.

The formula to calculate the frictional force is as follows:

$$F_f = \mu_s N$$

Here, coefficient of static friction is

μ_s

and normal force is

N

.

Step: 1

Draw the free body diagram of the crate A as shown in Figure (1).

C

Here, friction force acting on the crate A is

$$f_A$$

, mass of the crate is

$$m_A$$

, acceleration due to gravity is g and tension in the cable is T .

Apply the principle of equilibrium of forces along the y direction

$$(F_y)$$

.

$$\sum F_y = 0$$

$$N_A - m_A g = 0$$

$$N_A = m_A g$$

Substitute 50 kg for

$$m_A$$

and

$$9.81 \text{ m/s}^2$$

for g .

$$N_A = (50 \text{ kg}) \times (9.81 \text{ m/s}^2)$$

$$= 490.5 \text{ N}$$

Calculate the friction force in the block A.

$$f_A = \mu_s N_A$$

Substitute 0.25 for

$$\mu_s$$

and 490.5 N for

$$N_A$$

.

$$f_A = 0.25 \times 490.5\text{N}$$

$$= 122.62\text{N}$$

Apply the equilibrium of force along the x direction

$$(F_x)$$

.

$$\sum F_x = 0$$

$$T - f_A = 0$$

$$T = f_A$$

Substitute 122.625 N for

$$N_A$$

.

$$T = 0.25 \times 490.5\text{N}$$

$$T = 122.62\text{N}$$

Explanation:

Apply the principle of equilibrium along the y direction to find out the normal force exert on a crate A and using that calculated normal force in the friction force formula, find out the static friction force in the crate A. Finally apply the principle of equilibrium along the x direction to find out the tension in the cable.

Hint:

Determine the expression for normal force exert on crate B by applying the principle of equilibrium along the y direction and then determine the maximum force P that can be applied with causing the two crates to move applying the principle of equilibrium along the x direction.

Step: 2

Draw the free body diagram of the crate B as shown in Figure (2).

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Here, the friction force acting on the crate B is

$$f_B$$

, mass of the crate B is

$$m_B$$

, acceleration due to gravity is g , and load applied on the crate is P .

Apply the principle of equilibrium of forces along the y direction

$$(F_y)$$

.

$$\sum F_y = 0$$

$$N_B - m_B g + P \sin \theta = 0$$

$$N_B = m_B g - P \sin \theta$$

Substitute 50 kg for

$$m_B$$

,

$$9.81 \text{ m/s}^2$$

for g , and

$$30^\circ$$

for

$$\theta$$

.

$$N_B = (50 \text{ kg} \times 9.81 \text{ m/s}^2) - (P \times \sin(30^\circ))$$

$$= 490.5 - 0.5P$$

Calculate the friction force in the block B .

$$f_B = \mu_s N_B$$

Substitute 0.25 for

$$\mu_s$$

and

$$(490.5 - 0.5P)\text{N}$$

for

$$N_B$$

.

$$f_B = 0.25 \times (490.5 - 0.5P)$$

$$f_B = 122.62 - 0.125P$$

..... (1)

Apply the equilibrium of force along the x direction

$$(F_x)$$

.

$$\sum F_x = 0$$

$$-T - f_B + P \cos \theta = 0$$

Substitute

$$122.62\text{N}$$

for T and

$$30^\circ$$

for

$$\theta$$

.

$$-122.62\text{N} - f_B + P \cos (30^\circ) = 0$$

..... (2)

Substitute Equation (1) in Equation (2).

$$-122.62\text{N} - (122.62 - 0.125P) + P \cos(30^\circ) = 0$$

$$-122.62 - 122.62 + 0.125P + 0.866P = 0$$

$$0.995P = 245.24$$

$$P = 246.47\text{N}$$

Explanation:

Apply the principle of equilibrium along the y direction to find out the expression for the normal force exerted on the crate B and use that normal force expression in the friction force formula to find out the friction force in term of maximum load P . Finally apply the principle of equilibrium along the x direction to find out the maximum load that can be applied without causing the two 50 kg crates to move.

Answer:

The maximum force

(P)

that can be applied without causing the two crates to move is 246.47 N.